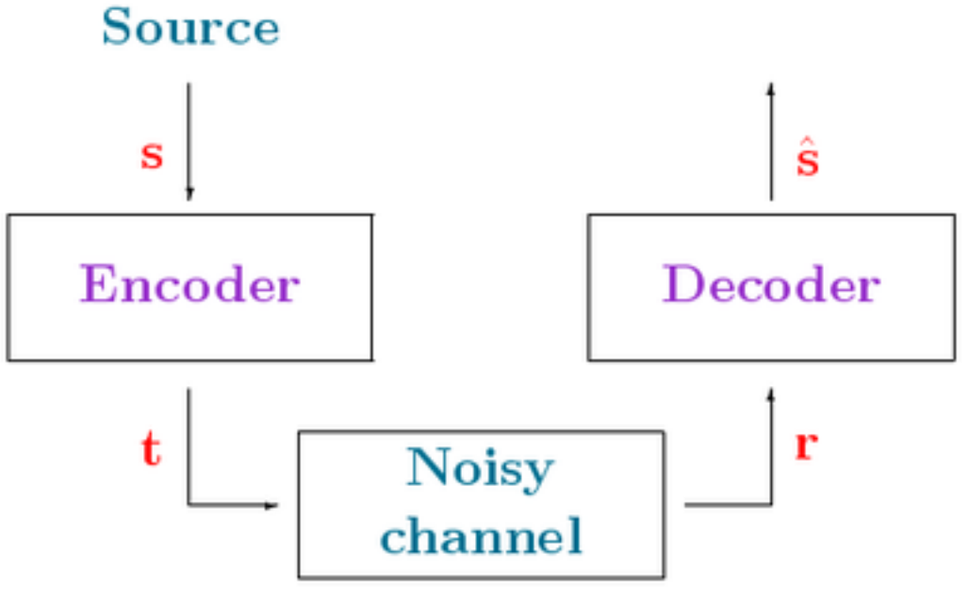
Data Compression

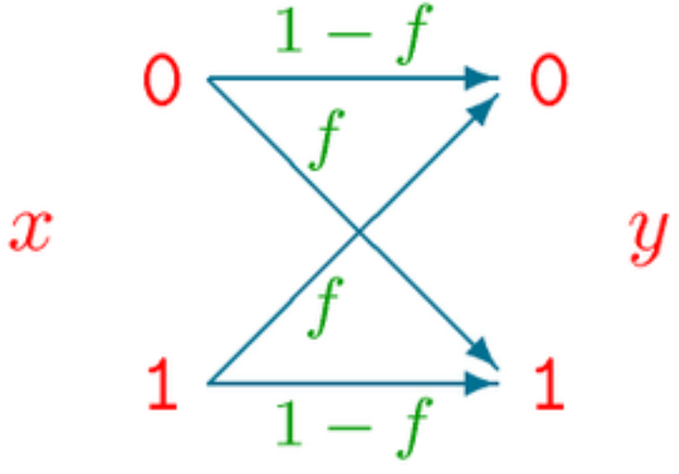
How to measure information content



adds redundancy

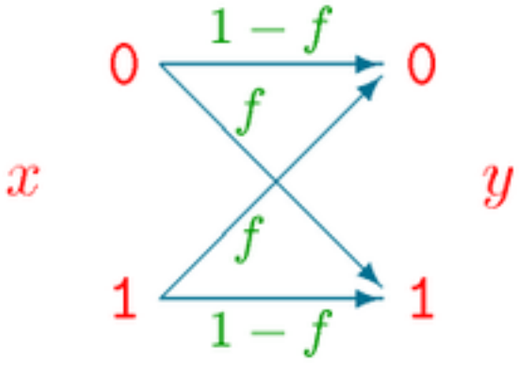
does inference

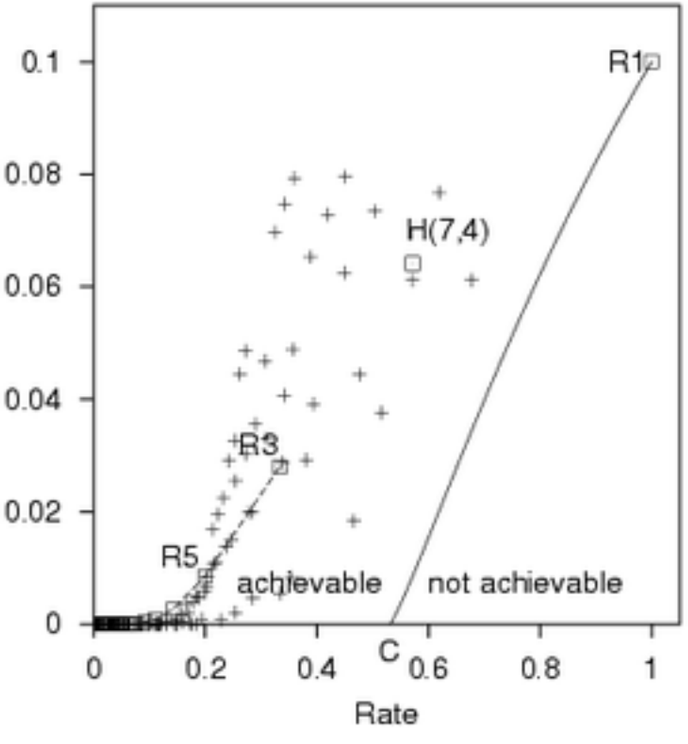
The binary symmetric channel



- Repetition codes
- (7,4) Hamming code

Shannon's noisy channel coding theorem





For any channel:

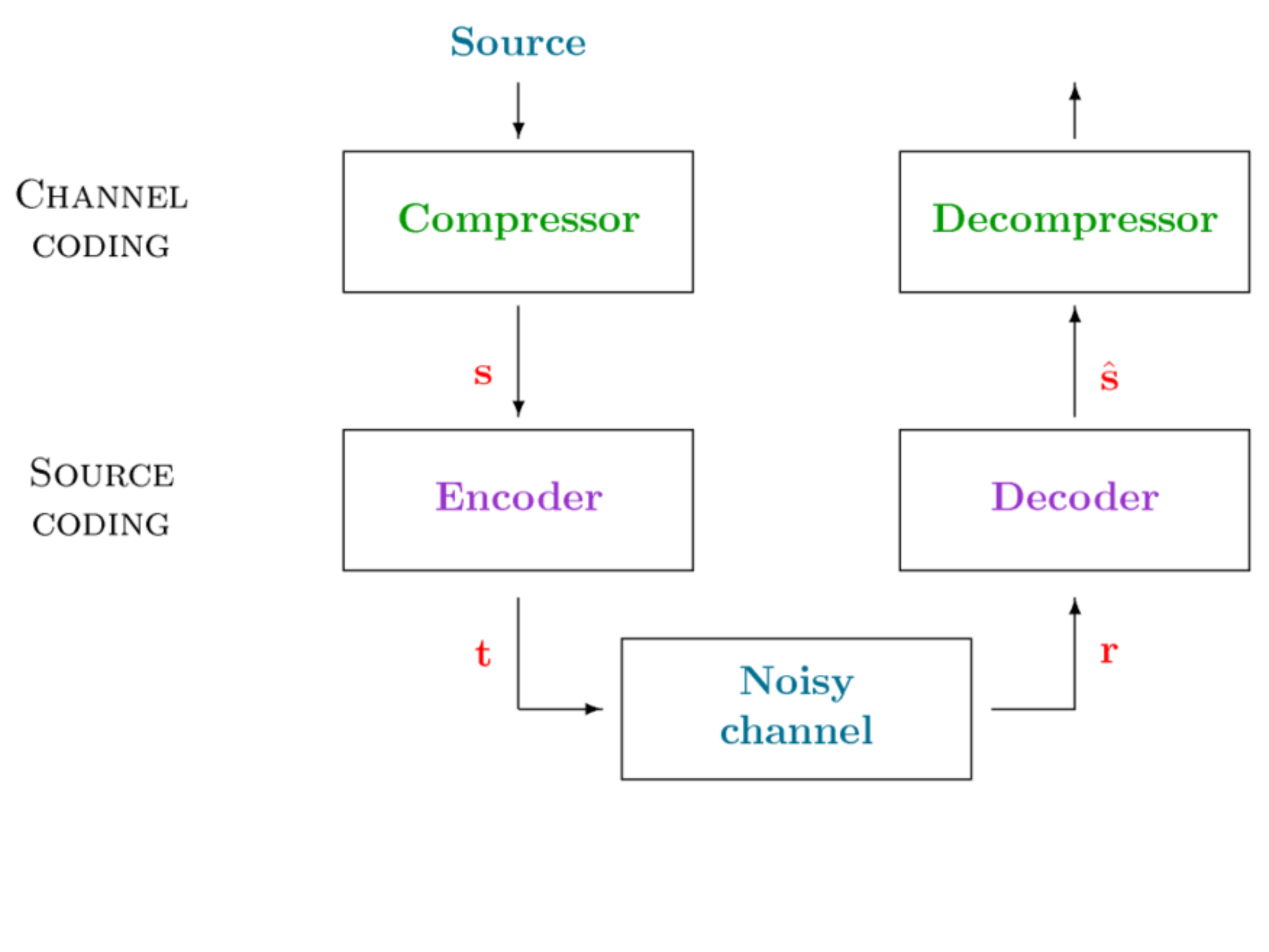
Reliable (virtually error-free) communication is possible

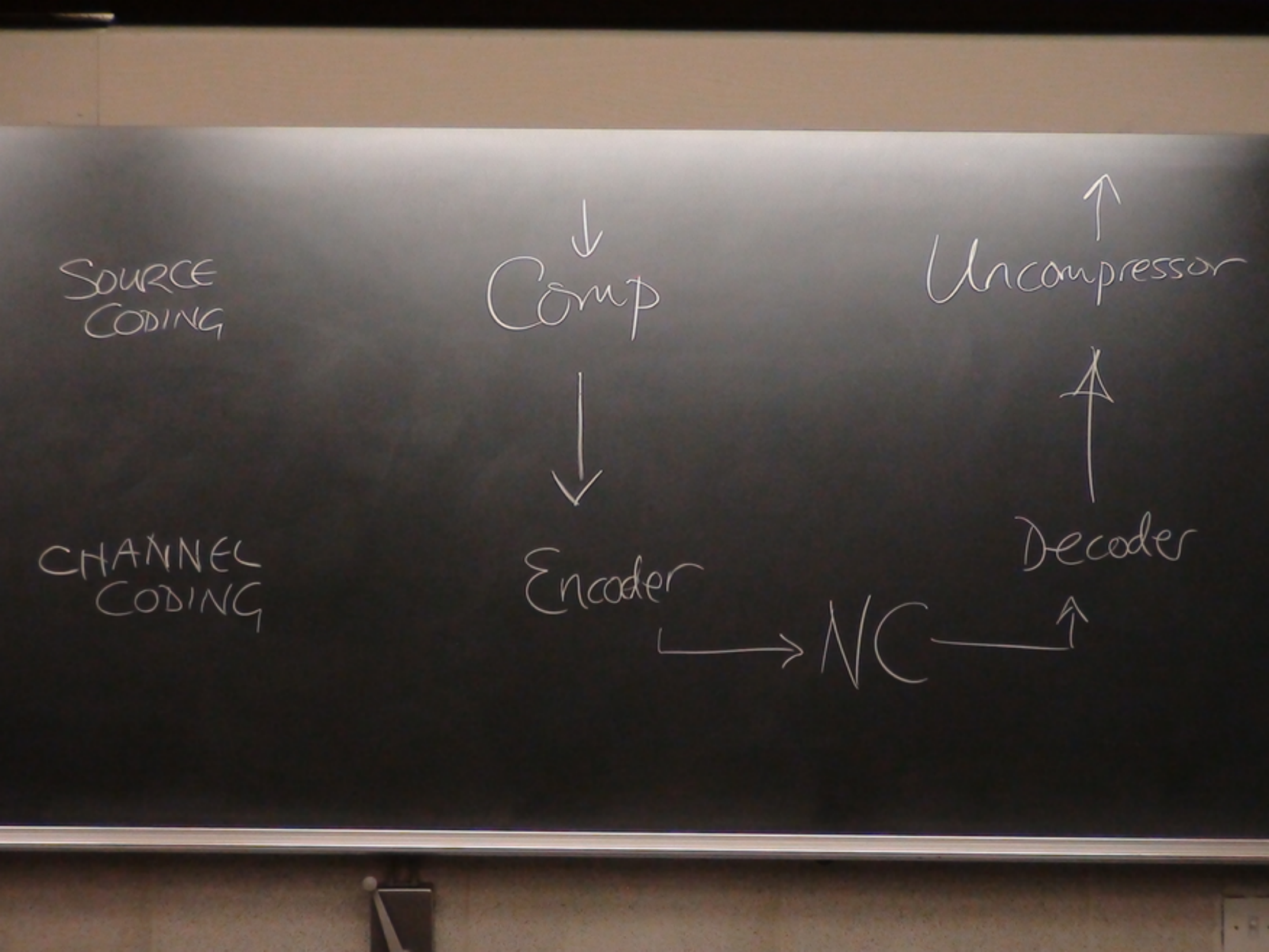
at rates up to C

Emma Woodh*use, hands*me, clever* and rich,*with a comfortab*e home an* happy di*position,*seemed to*unite som* of the b*st bless*ngs of e*istence;*and had *ived nea*ly twenty *ne year* in the*world w*th very*little *o distr*ss or vex*her. *he was*the yo*ngest *f the *wo dau*hters *f a most *ffect*onate* indu*gent *ather* and *ad, i* cons*quenc* of h*r si*ter'* mar*iage* bee* mis*ress*of h*s ho*se f*om a ver* ea*ly *eri*d. *er *oth*r h*d d*ed *oo *ong*ago*for*her to*ha*e *or* t*an*an*in*is*in*t *em*mb*an*e *f *er*ca*es*es* a*d*h*r*p*a*e*h*d*b*e* *u*p*i*d*b* *n*e*c*l*e*t*w*m*n*a* g**e**e**,**h**h** **l**n**i**l**s**r**o**a**o**e**i* a***c***n***S***e***y***s***d***s***a***r***e***n*** W****o****s****i****l****a****g****n****t****a****e****v***

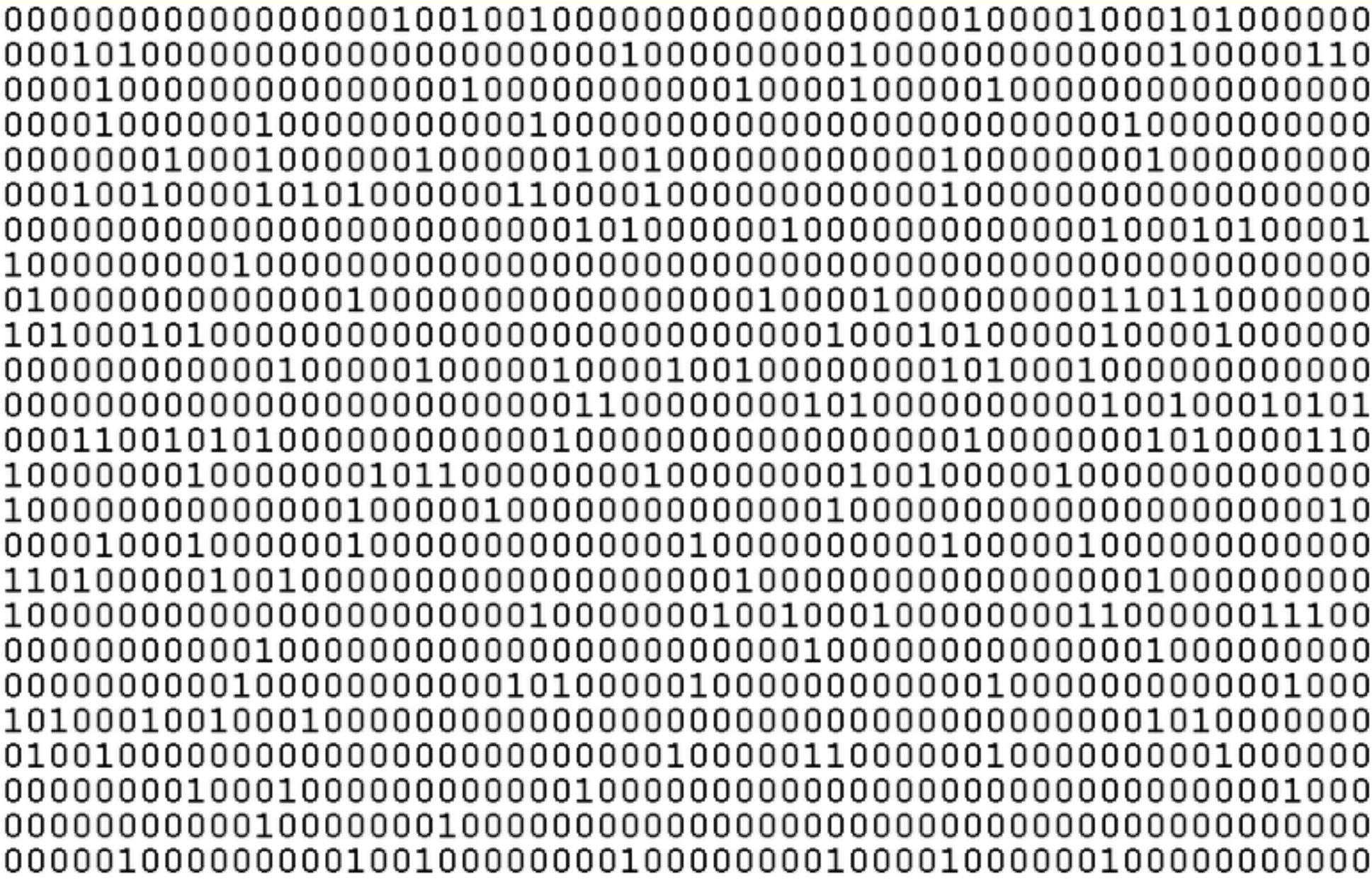
Emma Woodhouse, handsome, clever, and rich, with a comfortable home and happy disposition, seemed to unite some of the best blessings of existence; and had lived nearly twenty one years in the world with very little to distress or vex her. She was the youngest of the two daughters of a most affectionate, indulgent father; and had, in consequence of her sister's marriage, been mistress of his house from a very early period. Her mother had died too long ago for her to have more than an indistinct remembrance of her caresses; and her place had been supplied by an excellent woman as governess, who had fallen little short of a mother in affection. Sixteen years had Miss Taylor been in Mr Woodhouse's family, less as a governess than a friend, very

Emma Woodh*use, hands*me, clever* and rich,*with a comfortab*e home an* happy di*position,*seemed to*unite som* of the b*st bless*ngs of e*istence;*and had *ived nea*ly twenty *ne year* in the*world w*th very*little *o distr*ss or vex*her. *he was*the yo*ngest *f the *wo dau*hters *f a most *ffect*onate* indu*gent *ather* and *ad, i* cons*quenc* of h*r si*ter'* mar*iage* bee* mis*ress*of h*s ho*se f*om a ver* ea*ly *eri*d. *er *oth*r h*d d*ed *oo *ong*ago*for*her to*ha*e *or* t*an*an*in*is*in*t *em*mb*an*e *f *er*ca*es*es* a*d*h*r*p*a*e*h*d*b*e* *u*p*i*d*b* *n*e*c*l*e*t*w*m*n*a* g**e**e**,**h**h** **l**n**i**l**s**r**o**a**o**e**i* a***c***n***S***e***y***s***d***s***a***r***e***n*** W****o****s****i****l****a****g****n****t****a****e****v***


Source
Channel coding
Compressor
Decompressor
s
ŝ
Source coding
Encoder
Decoder
t
Noisy channel
r


SOURCE CODING
Comp
Uncompressor
CHANNEL CODING
Encoder
Decoder
NC

A simple redundant source - a bent coin



$p_1 = 0.1$

Bent coin

outcome $x \in \{ \text{Tails}, \text{Heads} \}$

probabilities $\{ 0.9, 0.1 \}$

$$P(x = \text{Tails}) = 0.9$$

such that

$$P(x = a_i) = p_i$$

$$\sum_i p_i = 1 \qquad p_i \geq 0$$

$p_1 = 0.1$

An "Ensemble" X is $\{x, A_x, P_x\}$

x: a random variable

A_x: a set of possible outcomes

$A_x = \{a_1, a_2, a_3, \ldots a_I\}$

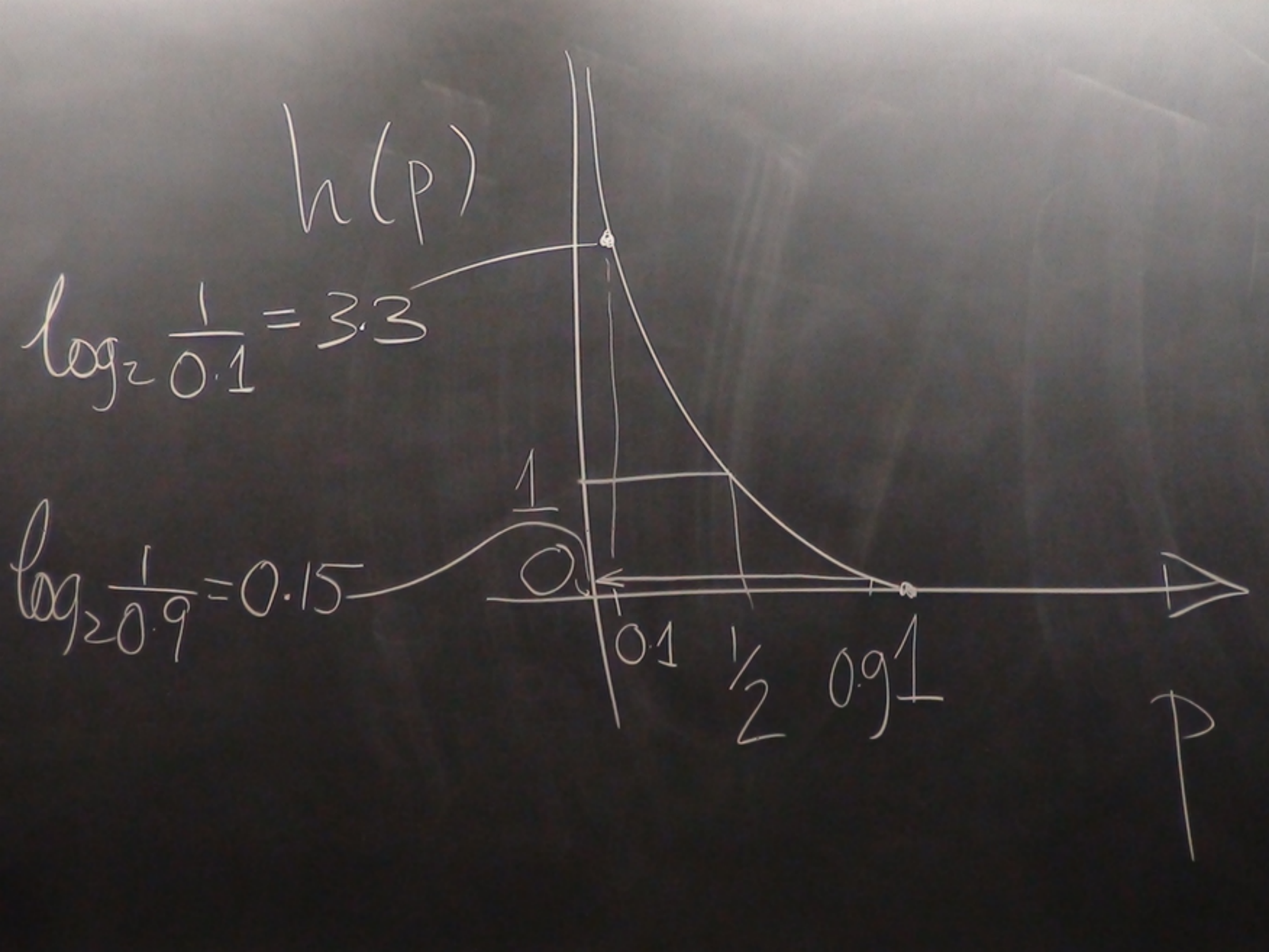
& a set of probabilities $P_x = \{p_1, p_2, \ldots p_I\}$

Idea

Shannon information content of an outcome

$$x = a_i$$

$$h(x=a_i) = \log_2 \frac{1}{P(x=a_i)} \text{ bits}$$


h(p)
$\log_2 \frac{1}{0.1} = 3.3$
$\log_2 \frac{1}{0.9} = 0.15$
1
0
0.1
1/2
0.9
1
p

$h(x)$ is the compressed file length to which we should aspire

Note h is additive for independent random variables

eg

x, y

$$P(x,y) = P(x)P(y)$$

$$\hat{h}(x,y) = \log_2 \frac{1}{P(x,y)}$$

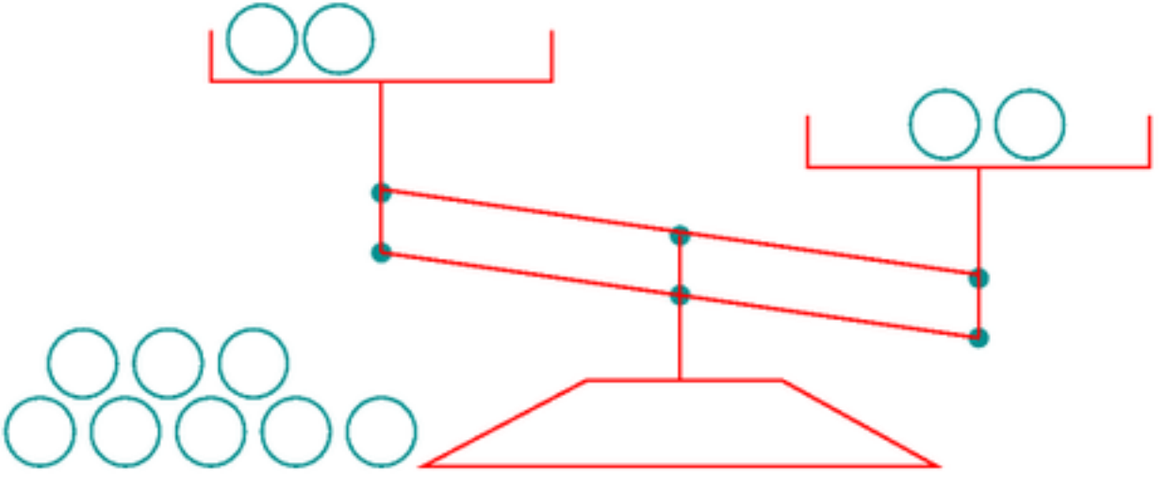
$$= \log_2 \frac{1}{P(x)} + \log_2 \frac{1}{P(y)}$$

Definition:

The ENTROPY of an ensemble is
the average Shannon information content

$$H(X) = \sum_x P(x) \log_2 \frac{1}{P(x)} \quad (\text{bits})$$

The weighing problem



You are given 12 balls, all equal in weight except for one that is either heavier or lighter.

Design a strategy to determine

which is the odd ball

and whether it is heavier or lighter,

in as few uses of the balance as possible.

My method needs 13 weighings

at most

12

11

10

9

8

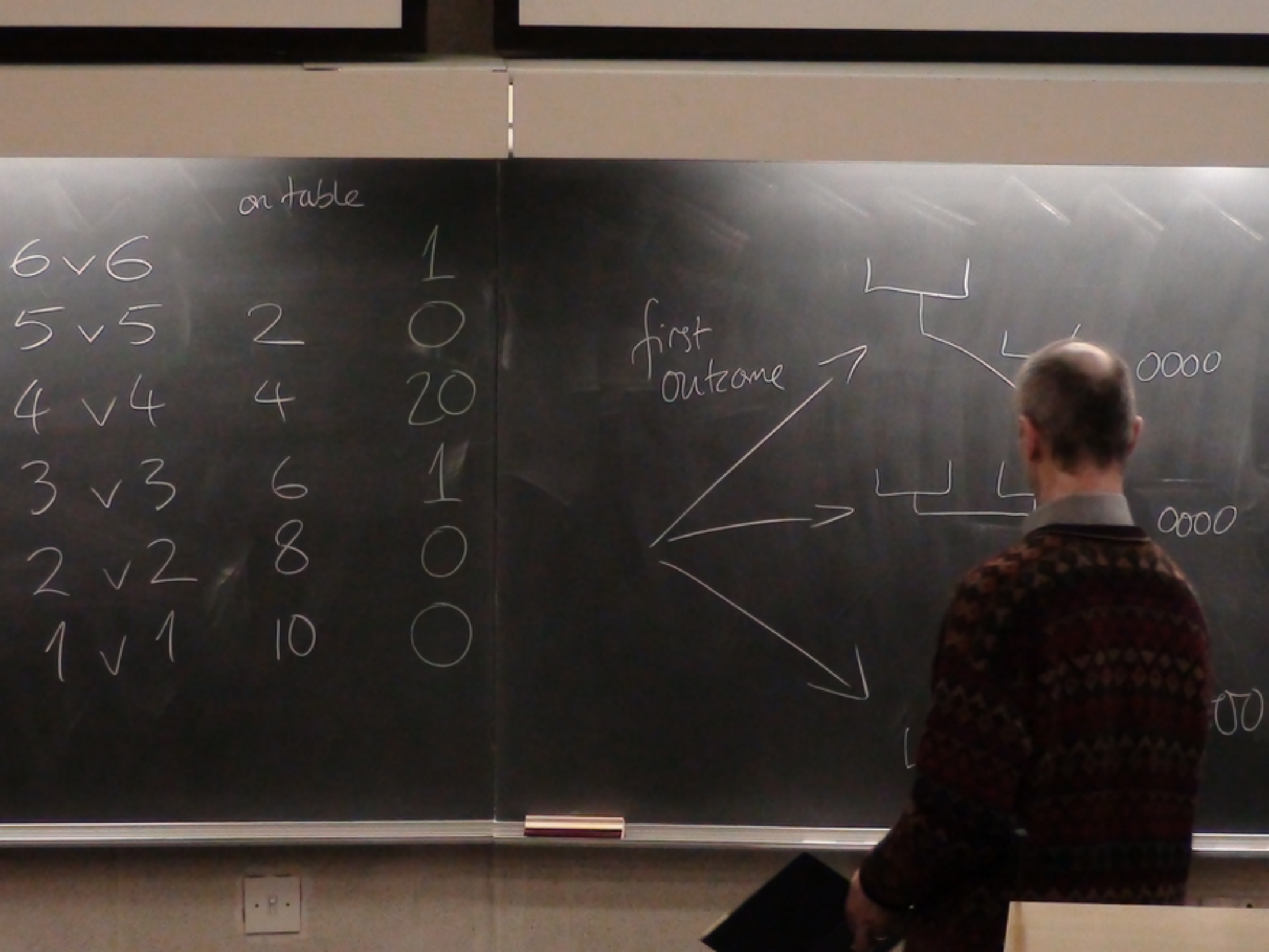
7 2

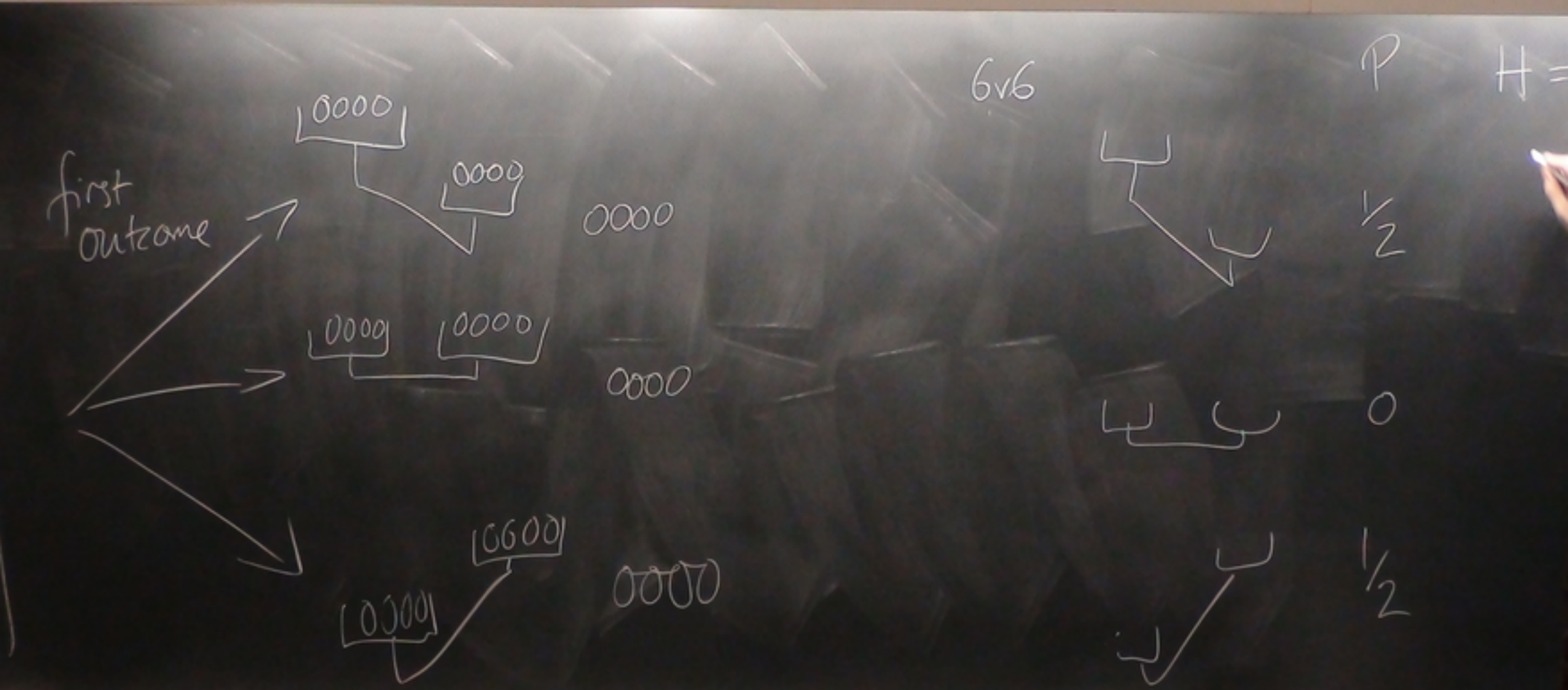
6

5 3

4 1

3 3


on table
6v6 1
5v5 2 0
4v4 4 20
3v3 6 1
2v2 8 0
1v1 10 0
first outcome
0000
0000
00


6v6
P
H =
first outcome
0000
0000
0000
0000
0000
0000
0000
0000
0000
0000
1/2
0
1/2

6v6

P

H = 1 bit

5v5

1/2

0

2

1/2

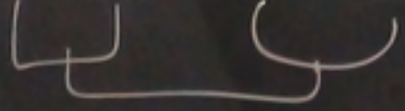
v6

P H = 1 bit

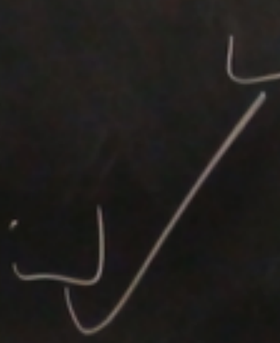
5v5

4v4

1/2 5/12



0 1/6 = 2/12 (2 on table)



1/2 5/12

6v6 H = 1 bit

5v5 H = 1.48 bits

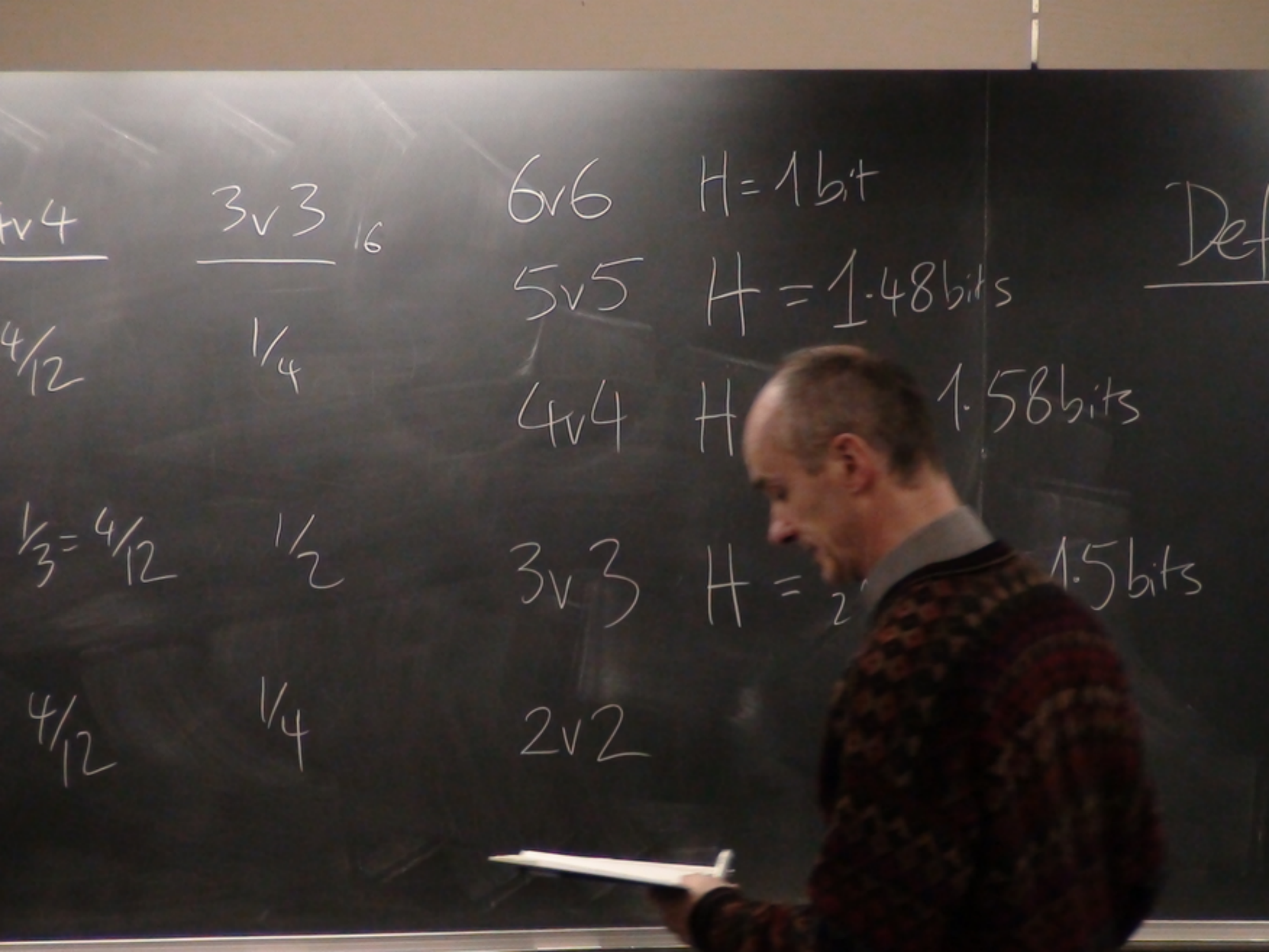
4v4

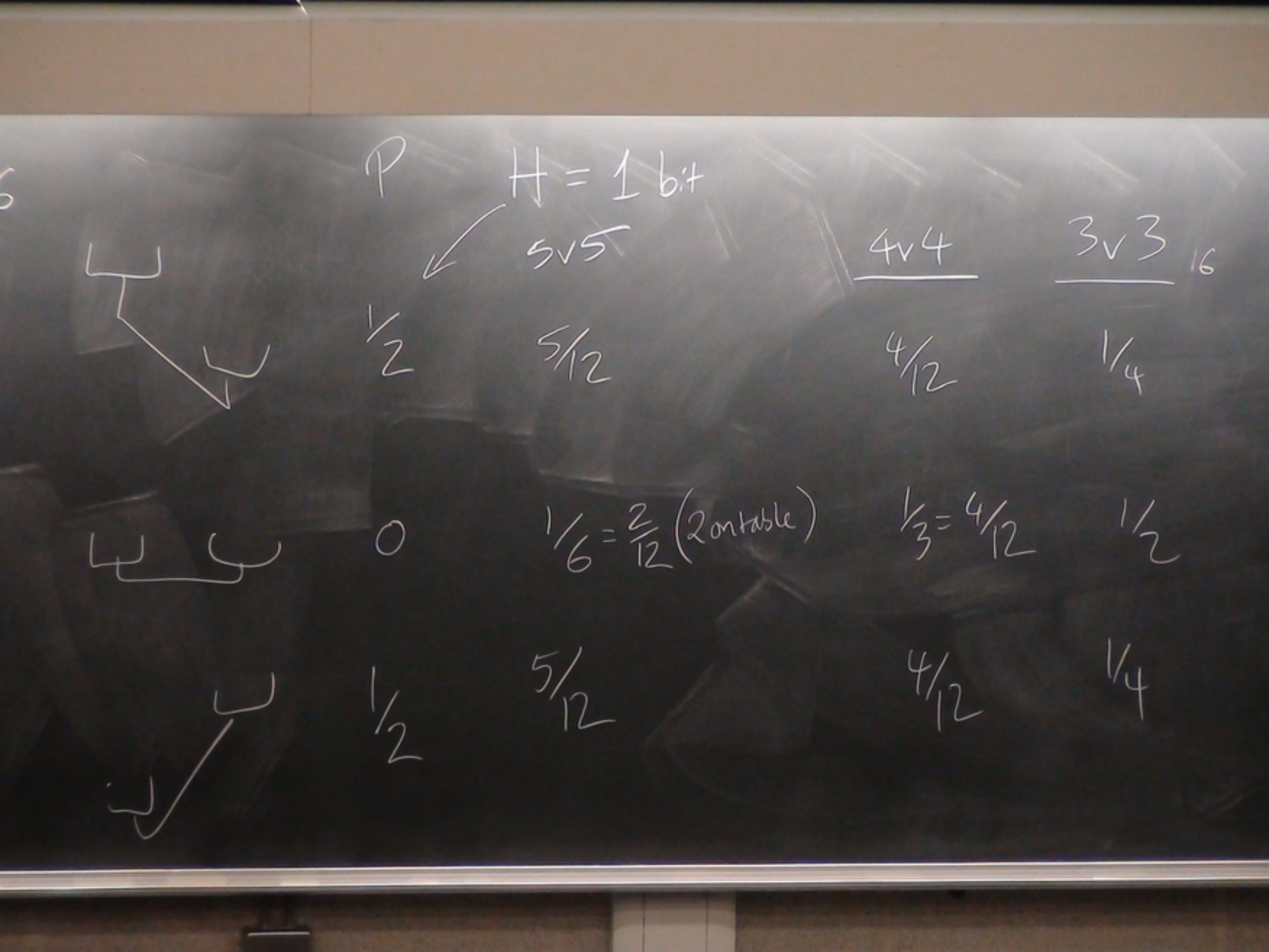
4v4

4/12

ble) 1/3 = 4/12

4/12


4v4
3v3
16
4/12
1/4
1/3 = 4/12
1/2
4/12
1/4
6v6 H = 1 bit
5v5 H = 1.48 bits
4v4 H
1.58 bits
3v3 H =
1.5 bits
2v2
Def


P
H = 1 bit
5v5
4v4
3v3
16
1/2
5/12
4/12
1/4
0
1/6 = 2/12 (2 on table)
1/3 = 4/12
1/2
1/2
5/12
4/12
1/4

3v3 16

1/4

6v6 H = 1 bit

5v5 H = 1.48 bits

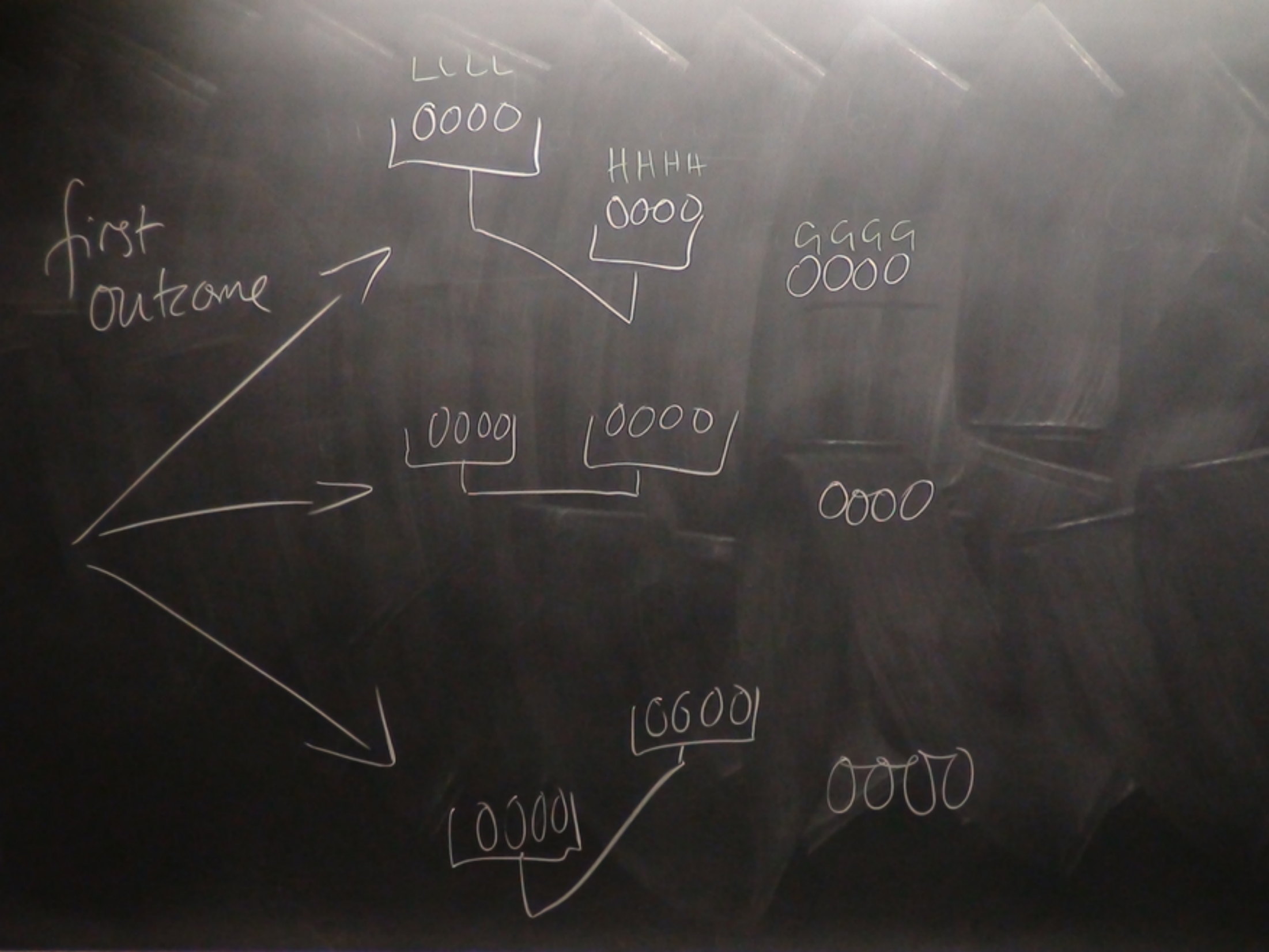
4v4 H = $\log_2$ = 1.58 bits

1/2 1/2 3v3 H = $\frac{1}{2} \times 2 + \frac{1}{2} \times 1$ = 1.5 bits

1/4 2v2 1.25

1v1 0.82

De


LLLL
0000
HHHH
0000
GGGG
0000
first
outcome
0000
0000
0000
0000
0000
0000

second weighing

GGGG v LLLL

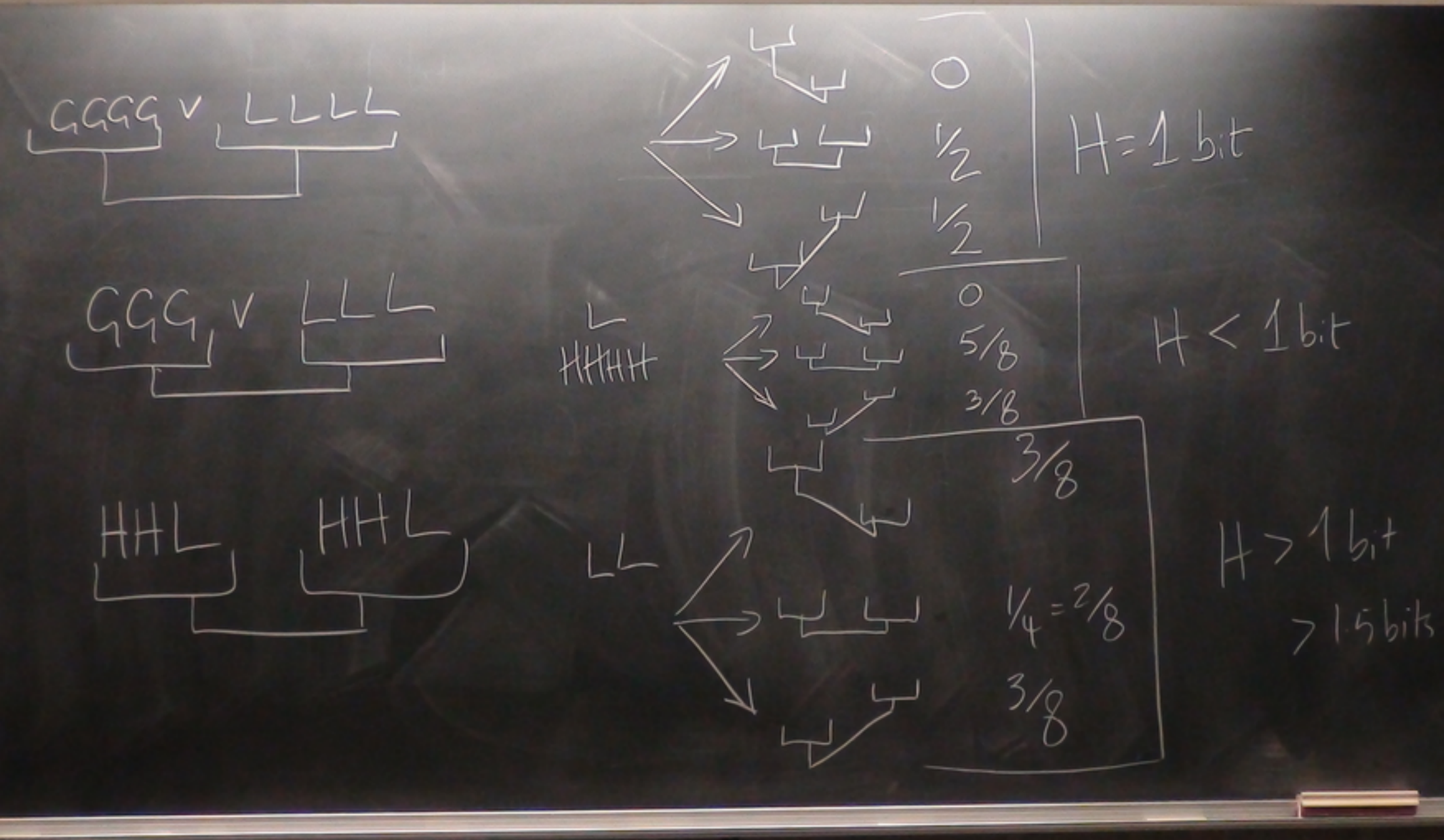
GGG v LLL

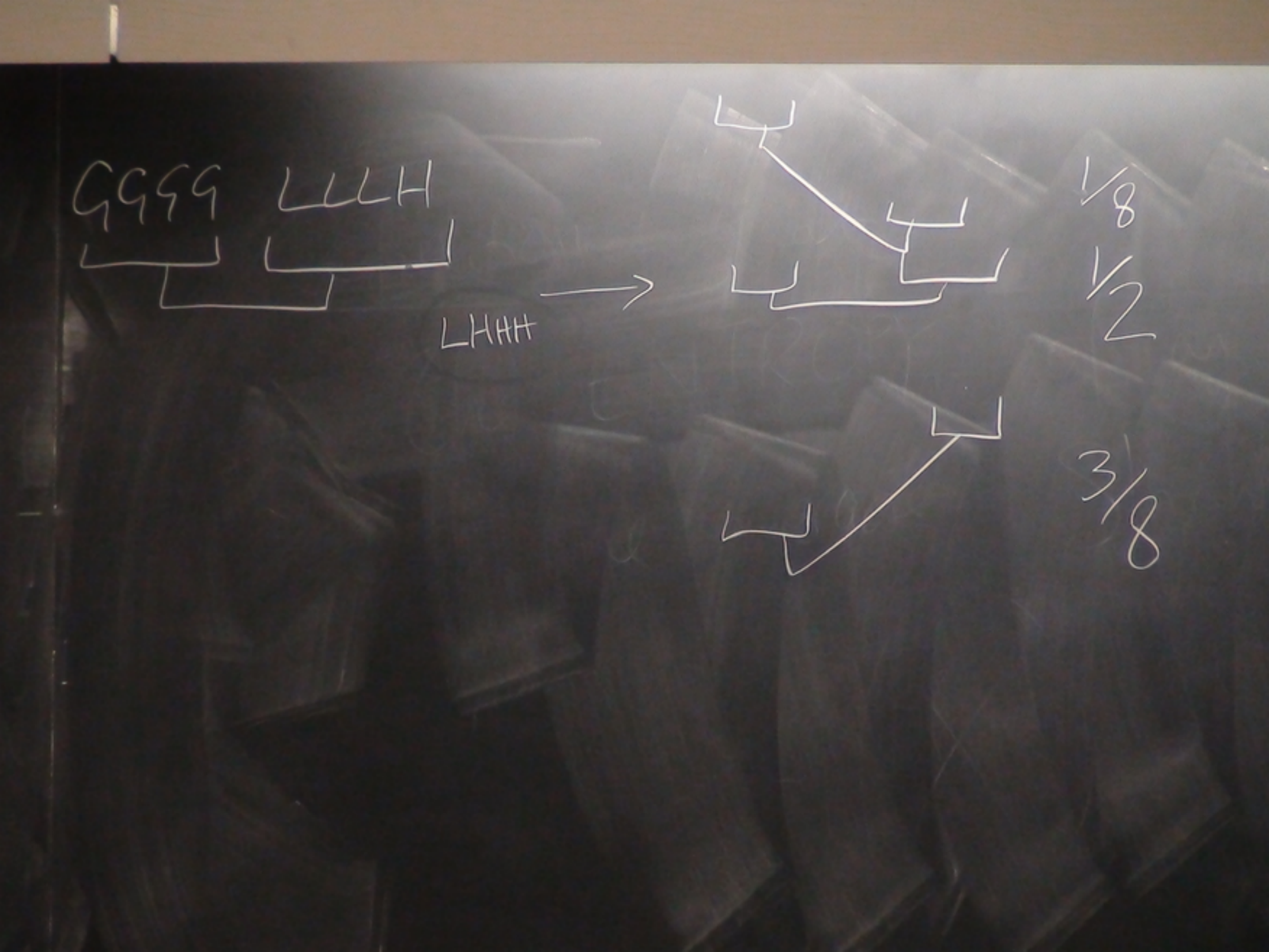
L

HHHH

HHL HHL

LL


GGGG v LLLL
0
1/2
1/2
H = 1 bit
GGG v LLL
L
HHHH
0
5/8
3/8
H < 1 bit
HHL
HHL
LL
3/8
1/4 = 2/8
3/8
H > 1 bit
> 1.5 bits


GGGG LLLH
LHHH
1/8
1/2
3/8

I $P = \{p_1, p_2 \ldots\ldots - p_I\}$

$H(X)$

what P maximizes $H(X)$?

$I \; P = \{p_1, p_2 \ldots\ldots - p_I\}$

$H(X)$

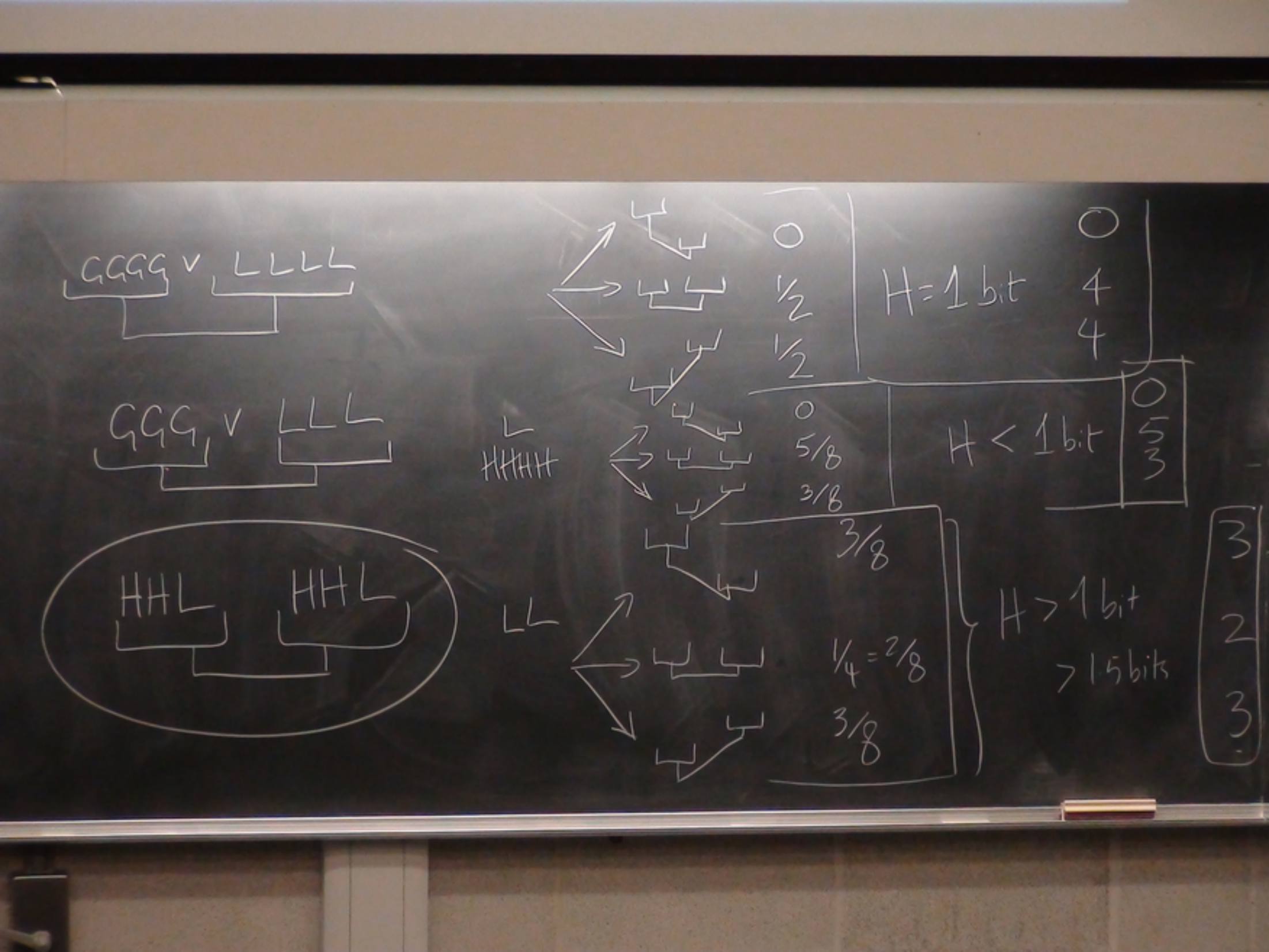
what $\vec{p}$ maximizes $H(X)$?

$$\sum p_i \log_2 \frac{1}{p_i}$$

$$p_i = \frac{1}{I}$$

8 pieces

3 children


GGGG v LLLL
0
1/2
1/2
H = 1 bit
0
4
4
GGG v LLL
L
HHHH
0
5/8
3/8
H < 1 bit
0
5
3
HHL HHL
LL
3/8
1/4 = 2/8
3/8
H > 1 bit
> 1.5 bits
3
2
3

The Bent Coin Lottery

A coin with $p_1 = 0.1$ will be tossed $N = 1000$ times.

The outcome is $\mathbf{x} = x_1 x_2 \ldots x_N$.

e.g., $\mathbf{x}$ = 000001001000100...00010

You can buy any of the 2^N possible tickets for £1 each, before the coin-tossing.

If you own ticket $\mathbf{x}$, you win £1,000,000,000.

Q To have a 99% chance of winning, at lowest possible cost, which tickets would you buy?

- And how many tickets is that?

  Express your answer in the form $2^{(\cdots)}$.

Please think about this question before lecture 3

Lottery tickets available

2^N

0000000000.....00000
0000000000.....00001
0000000000.....00010
0000000000.....00011
0000000000.....00100
0000000000.....00101
0000000000.....00110
0000000000.....00111
⋮
0010000001.....01000
⋮
1111111111.....11101
1111111111.....11110
1111111111.....11111

Homework

(Exercise 1.9, page 14)

Design an error-correcting code for the binary symmetric channel (or another noisy channel, you choose); invent a decoding algorithm for it, and estimate its probability of error. Be creative!

Other recommended exercises: 1.3, 1.5-7, 1.11; Reading: Chapters 1, 2, 4.